

Chapter 3

Homework 4 due: Tuesday 25th March, 2014

All problems without a ★ from section 3.2 and 3.3; exercises 3.14-3.19.

Exercise 3.14. Find the distinct ideals of $\mathbb{Z}_{12}$.

Proof. All of the ideals of $\mathbb{Z}_{12}$ are principal. It is easy to directly check that the distinct ideals of $\mathbb{Z}_{12}$ are:

$$\begin{aligned}\langle 0 \rangle &= \{0\} \\ \langle 1 \rangle &= \{1, 2, 3, 4, \dots, 11, 0\} \\ \langle 2 \rangle &= \{2, 4, 6, 8, 10, 0\} \\ \langle 3 \rangle &= \{3, 6, 9, 0\} \\ \langle 4 \rangle &= \{4, 8, 0\} \\ \langle 6 \rangle &= \{6, 0\}\end{aligned}$$

In general, the distinct ideals of $\mathbb{Z}_n$ will be all the divisors of n and $\langle 0 \rangle$. To see this suppose that $\gcd(n, m) = 1$. Then $\langle m \rangle = \{m, 2m, 3m, \dots, km\}$. Since n and m are relatively prime, $m(k-1) \not\equiv 0 \pmod{n}$, and $k = n$ since $|\langle m \rangle| \leq n$. So $\gcd(m, n) = 1$ implies that $\langle m \rangle = \langle 1 \rangle$.

Suppose $\text{lcm}(m, n) = k$. Then $\langle m \rangle = \{m, 2m, \dots, km\}$ where $m(k-1) \not\equiv 0 \pmod{n}$ since then $n|k-1$ and $k-1$ would be the lowest common multiple. Also $n \nmid (am - bm)$ for any distinct $0 \leq a, b \leq k$ since then $m(a-b)$ would be the lowest common multiple not k . So $\langle m \rangle$ has k distinct elements, and it follows that the distinct ideals of $\mathbb{Z}_n$ are the distinct divisors of n and 0. $\square$

Exercise 3.15. ★

Exercise 3.16. ★

Exercise 3.17. elements of $R/0_R$ and R/R .

Proof. $(R/0_R)$: Let $a + 0_R \in R/0_R$. For any $r \in R$, $a \equiv r \pmod{0_R}$ iff $a - r \in 0_R \iff a = r$. So then $a + 0_R = a$ for any $a \in R$. Recall that $R/0_R = R$.

(R/R) : Let $a + R \in R/R$. For any $r \in R$, $a \equiv r \pmod{R}$ iff $a - r \in R$. R is a ring so then $a - r \in R$ for all a and $r \in R$. This is true for 0_R , i.e., $a - 0_R \in R$ implying that $a + R \equiv 0 + R$ for all $a \in R$. Thus R/R has only 1 coset, $0 + R = R$. Recall that $R/R = 0$. $\square$

Exercise 3.18. with odd denominators, and let E be the set of elements in D with even numerators.

1. Prove that E is an ideal in D .
2. Prove that D/E has exactly two distinct cosets.

Proof. Part 1: Note that any two rational numbers can be written so that they have the same denominators. Let $\frac{2x}{2n+1}, \frac{2y}{2n+1} \in E$ and $\frac{r}{2n+1} \in D$ for integers $x, y, n, r \in \mathbb{Z}$.
 $\frac{2x}{2n+1} - \frac{2y}{2n+1} = \frac{2(x-y)}{2n+1}$ and $\frac{2x}{2n+1} \frac{r}{2n+1} = \frac{2xr}{4n^2+4n+1} = \frac{2(xr)}{2(2n^2+2n)+1} \in E$. So E is an ideal in D by the ideal test.

Part 2: Let $\frac{x}{2n+1} \in D$ then x is either even or odd. If it is even then $\frac{x}{2n+1} + E = 0 + E = E$ since $\frac{x}{2n+1} \in E$. If it is odd then we can write $\frac{2m+1}{2n+1}$ for some $m \in \mathbb{Z}$. Now let $\frac{2m'+1}{2n+1}$ for some $m' \in \mathbb{Z}$. $\frac{2m+1}{2n+1} - \frac{2m'+1}{2n+1} = \frac{2(m-m')}{2n+1}$ which has an even denominator thus $\frac{2m+1}{2n+1} + E = \frac{2m'+1}{2n+1} + E$. So there are exactly 2 distinct cosets. $\square$

- Note that since $1 = \frac{1}{1}, \frac{2m+1}{2n+1} + E = 1 + E$. So the two cosets can be written as $0 + E$ and $1 + E$.

Exercise 3.19. in $\mathbb{Z}[x]$.

1. Show that C is the principal ideal $\langle x \rangle$.
2. Show that $\mathbb{Z}[x]/C$ consists of an infinite number of distinct cosets, one for each $n \in \mathbb{Z}$.

Proof. Part 1: If $g(x) \in C$ then

$$g(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x = (x)(a_n x^{n-1} + a_{n-1} x^{n-2} + \cdots + a_1) = x \cdot f(x)$$

for some $f(x) \in \mathbb{Z}[x]$.

Part 2: Let $f_1(x) + C, f_2(x) + C \in \mathbb{Z}[x]/C$ for $f_1(x), f_2(x) \in \mathbb{Z}[x]$. From part one, $f_1(x) + C = f_1(x) + \langle x \rangle$ and $f_2(x) + C = f_2(x) + \langle x \rangle$. If $f_1(x) - f_2(x) \in \langle x \rangle$ then $f_1(x) - f_2(x)$ must have a zero constant term ($x \cdot g(x) = f_1(x) - f_2(x)$ for some $g(x) \in \mathbb{Z}[x]$).

$$\begin{aligned} f_1(x) - f_2(x) &= [a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0] - [b_n x^n + b_{n-1} x^{n-1} + \cdots + b_1 x + b_0] \\ &= (a_n - b_n)x^n + (a_{n-1} - b_{n-1})x^{n-1} + \cdots + (a_1 - b_1)x + (a_0 - b_0) \\ &= (a_n - b_n)x^n + (a_{n-1} - b_{n-1})x^{n-1} + \cdots + (a_1 - b_1)x + 0 \end{aligned}$$

Thus $a_0 - b_0 = 0 \Rightarrow a_0 = b_0$. So $f_1(x)$ and $f_2(x)$ have the same constant term. So two cosets $f_1(x) + \langle x \rangle$ and $f_2(x) + \langle x \rangle$ are equivalent if and only if $f_1(x)$ and $f_2(x)$ have the same constant terms. These are polynomials with integer coefficients so there is a distinct coset for each integer. $\square$